UNIT 19 CAPITAL ASSET PRICING MODEL

Structure

19.0 OBJECTIVES

After completing this unit, you will be able to:

- explain the concept of CAPM;
- spell out the composition of Sharpe's theory;
- highlight the importance, role and functions of the CAPM model in an economy;
- identify assumptions of the CAPM model;
- describe the CAPM model;
- analyse the applications of the CAPM model and its empirical analysis; and
- indicate the problems of the CAPM model.

19.1 INTRODUCTION

You must have understood that financial markets play a very important role in the economy by facilitating the allocation of capital managing risk and enabling price discovery. They are influenced by various factors such as economic conditions, government policies, sentiment of investors and geopolitical events. A proper understanding and analysis of these markets are necessary for investors, policymakers and businesses to make informed decisions regarding investment, risk management and capital allocation. We have already discussed about Efficient Portfolio frontier and Markowitz's Portfolio theory in Unit 18. We have also discussed the relationship between risk and return in Unit 18. We will extend our understanding of risk, return and variance in this unit and discuss the Capital Asset Pricing Model (CAPM).

Models like CAPM provide valuable insights into the pricing and behaviour of these financial markets. CAPM provides valuable insight into the relationship between risk and return and financial markets. This unit will explore the foundational principles of CAPM and will also help in examining the implications for the economy as a whole. This unit will also help the learners in analyzing the role of CAPM in asset pricing, portfolio management and decision-making. This unit will also throw light on how CAPM influences economic outcomes and market efficiency. The limitations and critiques of CAPM will also be discussed.

19.2 THE CAPITAL ASSET PRICING MODEL (CAPM)

The Capital Asset Pricing Model (CAPM) is a fundamental framework in finance that plays a crucial role in asset pricing, portfolio management, and investment decision-making. Developed in the 1960s by William Sharpe, John Linter, and Jan Mossin, CAPM provides a theoretical foundation for understanding the relationship between risk and return in financial markets. Its significance in finance stems from its ability to quantify the expected return on an investment based on its level of systematic risk, offering valuable insights for investors, analysts, and policymakers.

Before delving into the key aspects of the CAPM model, let us discuss some key concepts that form the basis of CAPM. These are as follows:

Systematic Risk: We have read about systematic and non-systematic risk in Unit 18. Systematic risk, also known as market risk, refers to the risk that cannot be diversified away by holding a diversified portfolio. It is associated with factors that affect the entire market, such as economic downturns, interest rate changes, or geopolitical events.

Beta: Beta measures the sensitivity of an asset's returns to changes in the overall market returns. It quantifies the extent to which an asset moves in tandem with the market. Assets with a beta of 1 move in perfect correlation with the market, while assets with a beta greater than 1 are more volatile, and those with a beta less than 1 are less volatile.

Risk-Free Rate: The risk-free rate represents the return on an investment with zero risk, typically provided by the yield on government bonds. It serves as the baseline return that investors can earn without taking on any investment risk.

Market Risk Premium: The market risk premium is the additional return that investors demand for bearing systematic risk. It is calculated as the difference between the expected return on the market portfolio and the risk-free rate.

Formulation of the CAPM Equation: The CAPM equation expresses the relationship between expected return, systematic risk (beta), and the risk-free rate:

Expected Return=R_f+β× (R_m−R_f)

Where:

- Expected Return: Expected Return is the expected return on the asset.
- R_f is the risk-free rate, representing the return on investment with zero risk, typically proxied by the yield on government bonds.
- β is the asset's beta, which measures its systematic risk or sensitivity to market movements.
- R_m is the expected return on the market portfolio, usually represented by a broad market index like the S&P 500.
- (R_m−R_f) is the market risk premium, representing the excess return investors demand for bearing the systematic risk of the market portfolio.

19.2.1 The Distinctive Features of the Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is an extremely valuable tool to analyse the financial market. Some of the distinctive features of the model are as follows:

1) **Systematic Risk Emphasis:** CAPM focuses on systematic risk, also known as non-diversifiable risk, which is the risk inherent to the entire market or market segment. It assumes that investors are primarily concerned with this type of risk because it cannot be eliminated through diversification.

2) **Single-Factor Model:** CAPM is a single-factor model, relying on beta as the sole measure of risk. Beta measures an asset's sensitivity to market movements, providing a straightforward way to assess its risk relative to

the market portfolio. This simplicity makes CAPM easy to understand and apply.

3) **Relationship between Risk and Return:** CAPM quantifies the relationship between an asset's risk and its expected return. It asserts that investors require higher returns for bearing higher levels of systematic risk, as reflected in the market risk premium.

4) **Market Portfolio Benchmark:** CAPM uses the market portfolio as a benchmark for risk and return. The market portfolio represents a diversified portfolio containing all risky assets in the market, weighted according to their market capitalization. By comparing an asset's expected return to the return on the market portfolio, investors can assess its performance and risk-adjusted return.

5) **Risk-Free Rate Incorporation:** CAPM incorporates the risk-free rate, representing the return on an investment with zero risk. This element allows investors to differentiate between the risk-free return and the additional return required for bearing systematic risk, facilitating investment decision-making.

6) **Equilibrium Pricing:** CAPM is based on the concept of market equilibrium, suggesting that asset prices adjust to reflect the relationship between risk and return implied by the model. In equilibrium, the expected returns of all assets align with their systematic risk as measured by beta.

7) **Universal Applicability:** CAPM's principles apply across various asset classes and markets, making it a widely used model for determining expected returns on stocks, bonds, and other financial instruments. Its broad applicability enhances its utility in investment analysis and portfolio management.

19.3 IMPORTANCE OF SHARPE'S THEORY

Sharpe ratio is one of the indexes derived from the CAPM model. It is used to evaluate the value of investment in a portfolio. This section discusses the objectives, functions and application of the Sharpe ratio.

19.3.1 Objectives and Functions of Sharpe's Theory

The Sharpe ratio is a measure of risk-adjusted return, commonly used in finance to assess the performance of an investment or portfolio relative to its risk. The formula for the Sharpe ratio is:

Sharpe ratio= $(R_p - R_f) / \sigma_p$

Where:

- R_p is the expected return on the investment or portfolio.

- R_f is the risk-free rate of return (typically the yield on government bonds).

- σ_p is the standard deviation of the investment or portfolio's returns, representing its volatility or risk.

- The numerator ($R_p - R_f$) calculates the excess return of the investment or portfolio over the risk-free rate, while the denominator σ_p measures its volatility or risk. Dividing the excess return by the risk provides a measure of risk-adjusted return, with a higher Sharpe ratio indicating better risk-adjusted performance.

Its primary objective and functions in the economy include:

i) **Objective Measurement of Risk-Adjusted Performance**: The main objective of the Sharpe ratio is to provide a standardized measure of risk-adjusted performance. It evaluates the excess return of an investment or portfolio relative to its risk, typically measured as the standard deviation of returns. By considering both return and risk, the Sharpe ratio allows investors to compare investments consistently.

ii) **Quantifying Risk-Return Tradeoff:** The Sharpe ratio quantifies the tradeoff between risk and return by expressing the excess return per unit of risk taken. A higher Sharpe ratio indicates better risk-adjusted performance, as it reflects higher returns relative to the volatility of those returns. This function helps investors evaluate whether an investment's return compensates adequately for the level of risk it entails.

iii) **Portfolio Allocation Decision:** In portfolio management, the Sharpe ratio assists investors in making allocation decisions by comparing the risk-adjusted returns of different asset classes or investment strategies. It enables investors to identify portfolios or assets that offer the highest return per unit of risk and optimize their portfolio allocations accordingly.

iv) **Performance Evaluation Tool**: The Sharpe ratio serves as a performance evaluation tool for investment managers, financial analysts, and fund selectors. It allows them to assess the skill of portfolio managers or the performance of investment funds relative to their benchmark or peer group. A higher Sharpe ratio suggests superior risk-adjusted performance, while a lower ratio may indicate underperformance or excessive risk-taking.

v) **Risk Management Tool:** The Sharpe ratio facilitates risk management by providing insights into the risk-adjusted returns of investment portfolios. Investors can use it to identify and manage portfolio risk exposures, ensure diversification, and optimize risk-return tradeoffs according to their risk preferences and investment objectives.

vi) **Benchmark Comparison:** The Sharpe ratio enables investors to compare the risk-adjusted performance of their portfolios or investments against a benchmark, such as a market index or a target return. This comparative analysis helps investors gauge whether their portfolio's returns are commensurate with the level of risk taken relative to a chosen benchmark.

Thus, the Sharpe ratio plays a crucial role in the economy by providing a standardized measure of risk-adjusted performance, aiding portfolio allocation decisions, evaluating investment performance, managing portfolio risk, and facilitating benchmark comparisons. Its objective function is to help investors make informed decisions by assessing the risk-return tradeoff of investments or portfolios.

19.3.2 Importance and Role of Sharpe's Theory in an Economy

While the Sharpe ratio is more commonly used in the realm of finance and investment, its principles can still be applied to macroeconomic analysis, particularly in evaluating the risk-adjusted performance of economic policies or strategies. Let's consider some hypothetical examples to illustrate the importance and role of the Sharpe ratio in macroeconomics:

i) **Evaluation of Fiscal Policy:** Suppose a government is considering implementing an expansionary fiscal policy to stimulate economic growth. They plan to increase government spending and reduce taxes to boost aggregate demand. However, this policy entails higher government debt and potential inflationary pressures. The Sharpe ratio can be used to assess the risk-adjusted effectiveness of this fiscal policy. We can calculate the ratio by comparing the expected increase in GDP (or other relevant macroeconomic indicators) to the associated increase in government debt or inflation volatility. If the Sharpe ratio for the proposed fiscal policy is high, it indicates that the potential benefits (e.g., higher GDP growth) outweigh the associated risks (e.g., increased debt or inflation). A low Sharpe ratio, on the other hand, suggests that the policy's risks may outweigh the benefits.

ii) **Analysis of Monetary Policy:** Consider a Central Bank implementing an accommodative monetary policy by lowering interest rates to stimulate investment and consumption. While this policy may spur economic activity, it also carries risks such as asset price bubbles or currency depreciation. The Sharpe ratio can help evaluate the risk-adjusted impact of monetary policy actions. By comparing the expected increase in economic output or employment to the volatility of asset prices or exchange rates, policymakers can gauge the effectiveness and prudence of their monetary policy decisions. A high Sharpe ratio would indicate that the benefits of the monetary policy (e.g., higher economic growth) outweigh the risks (e.g., asset bubbles). Conversely, a low

Sharpe ratio suggests that the policy's risks may outweigh the benefits, signalling the need for alternative policy measures.

iii) **Assessment of Structural Reforms:** Imagine a country implementing structural reforms aimed at improving long-term economic growth, such as deregulation, labour market reforms, or investment in infrastructure. While these reforms may yield significant economic benefits over time, they can also entail short-term disruptions or adjustment costs. The Sharpe ratio can be employed to assess the risk-adjusted effectiveness of structural reforms. By comparing the expected long-term benefits of the reforms to the short-term costs or uncertainties, policymakers can determine whether the reforms offer a favourable risk-return tradeoff. A high Sharpe ratio would suggest that the potential long-term benefits of the reforms justify the short-term costs or uncertainties. Conversely, a low Sharpe ratio may indicate that the risks associated with the reforms outweigh the expected benefits, prompting policymakers to reconsider their approach.

It is also important to note that while the application of the Sharpe ratio in macroeconomics may require adaptations and interpretations compared to its use in finance, it can still offer valuable insights into the risk-adjusted performance of economic policies and strategies. By quantifying the tradeoff between expected benefits and associated risks, policymakers can make more informed decisions to promote macroeconomic stability and sustainable growth.

19.4 APPLICATION OF CAPITAL ASSET PRICING MODEL

In financial economics, the Capital Asset Pricing Model (CAPM) serves as a fundamental tool for understanding the relationship between risk and return in financial markets. Here's how it's applied in financial economics, explained with an example:

Expected Return Estimation: CAPM is used to estimate the expected return of an asset or a portfolio based on its systematic risk, as measured by its beta (β). The formula for the expected return of an asset according to CAPM is:

$E(R_i) = R_f + \beta_i \times (E(R_m) - R_f)$

- Where:
- $E(R_i)$ is the expected return of asset i,
- R_f is the risk-free rate,
- β_i is the beta of asset i,
- $E(R_m)$ is the expected return of the market portfolio.

An example of this application could be estimating the expected return of a particular stock. Suppose the risk-free rate is 3%, the expected return of the market portfolio is 8%, and the stock's beta is 1.2. Using CAPM, we can calculate the expected return of the stock as follows:

=0.03+1.2×(0.08−0.03)=0.03+1.2×0.05=0.03+0.06=0.09E(Ri
=0.03+1.2×(0.08−0.03)=0.03+1.2×0.05=0.03+0.06=0.09

So, the expected return of the stock according to CAPM would be 9%.

Cost of Equity Calculation: CAPM is used to determine the cost of equity for a company, which represents the return required by investors for holding the company's stock. The cost of equity is a crucial input in various financial decisions, such as capital budgeting and valuation. Companies use CAPM to calculate their cost of equity by substituting the company's beta into the CAPM formula. For example, if a company has a beta of 1.5, the risk-free rate is 4%, and the market risk premium is 6%, then the company's cost of equity according to CAPM would be:

=0.04+1.5×0.06=0.04+0.09=0.13Cost of Equity=0.04+1.5×0.06=0.04+0.09=
0.13

So, the cost of equity for the company would be 13%.

Risk-Adjusted Performance Measurement: CAPM is used in financial economics to evaluate the performance of investment portfolios or funds. By comparing the actual returns of a portfolio with the returns predicted by CAPM based on the portfolio's risk (beta), investors can assess whether the portfolio manager has generated excess returns (alpha) or underperformed the market. For instance, if a portfolio's actual return is 10% and its expected return based on CAPM is 8%, the portfolio manager has generated an excess return of 2%, indicating positive alpha.

These are just a few examples of how CAPM is applied in financial economics to make informed decisions regarding asset pricing, cost of capital, and performance evaluation in financial markets.

Check Your Progress 1

1) Define Capital Asset Pricing Model.

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2) Describe the main features of the Capital Asset Pricing Model.

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